KKT Condetions

→ Now if we have equally type constraints then we use Lagrange function, Means Lagrange multiplier method to find out Optimal solution of that of problems.

→ But if we have a constraint Optimization with inequality type constraints, then we use

→ Karaush - Kuhn - Tucker Condetions to find ~~out~~ Optimal solutions of such problems.

KK Condetion

· Consider the following constrained Optimization Problem with inequality constraints:

$$(MP) \quad \text{Min } f(x)$$

$$\text{Subject to: } g_i(x) \leq 0, \quad i = 1 = 1, 2, \ldots m$$

where f and $g_i : R^n \to R$ are defined and Continously defferentiable functions.

Necessary Part: Let $\bar{x}$ be a local min point of the problem (MP) at which basic constraint qualification holds. Then there exist multipliers (called KKT multipliers) $\bar{\lambda}_i$, $i = 1, 2 \ldots m$ such that the following condetions hold:

① $\nabla f(\bar{x}) + \sum_{i=1}^{m} \bar{\lambda}_i \nabla g_i(\bar{x}) = 0,$

② $g_i(\bar{x}) \le 0, \quad i = 1, 2, \dots m$

③ $\bar{\lambda}_i g_i(\bar{x}) = 0, \quad i = 1, 2 \dots m,$

(4) $\bar{\lambda}_i \ge 0$ for all i.

These conditions are called KKT Conditions.

Sufficient Conditions:

Let $(\bar{x}, \bar{\lambda}_1, \bar{\lambda}_2, \dots \bar{\lambda}_m)$ satisfy the KKT- Conditions.
Let f and $g_i \; \forall i$ be differentiable convex functions.

Then $\bar{x}$ is a global min point of the Problem (MP).

Proof:

$$\text{Min } f(x)$$
$$\text{s.t } g_i(x) \le 0 \quad i = 1, 2, \dots m$$

function f is differentiable convex function. What does it means

$$f(x) - f(\bar{x}) \ge (x - \bar{x})^T \nabla f(\bar{x}), \; \forall \; x \in S \longrightarrow ①$$

where $S = \{ x \in \mathbb{R}^n : g_i(x) \le 0, \; i = 1, 2 \dots m \}$

$g_i(x)$ is also a convex function

$$g_i(x) - g_i(\bar{x}) \ge (x - \bar{x})^T \nabla g_i(\bar{x}), \; \forall \; x \in S$$

from 1 to m like necessary Condition point (3)
multiply by λ_i

$$\sum_{i=1}^{m} \lambda_i g_i(x) - \sum_{i=1}^{m} \lambda_i g_i(\bar{x}) \ge (x - \bar{x})^T \sum_{i=1}^{m} \lambda_i \nabla g_i(\bar{x}) \longrightarrow ②$$

Adding ① & ②

$$f(x) - f(\bar{x}) + \sum_{i=1}^{m} \lambda_i g_i(\bar{x}) - \sum_{i=1}^{m} \bar{\lambda}_i g_i(\bar{x}) \geq$$

$$\underline{(x-\bar{x})^T} \nabla f(\bar{x}) + \underline{(x-\bar{x})^T} \sum_{i=1}^{m} \lambda_i \nabla g_i(\bar{x})$$

$$\Rightarrow (x-\bar{x})^T \left[\nabla f(\bar{x}) + \cancel{\sum_{i=1}^{m} \lambda_i \nabla g_i(\bar{x})}\right]$$

0

↳ From necessary condt ①

∴

$$f(x) - f(\bar{x}) + \sum_{i=1}^{m} \lambda_i g_i(x) - \cancel{\sum_{i=1}^{m} \bar{\lambda}_i g_i(\bar{x})} \geq 0$$

↳ this part = 0

From necessary ③ $\lambda_i g_i(\bar{x}) = 0$ bec of this.

$$\therefore \quad f(x) - f(\bar{x}) + \sum_{i=1}^{m} \lambda_i g_i(x)$$

$g_i(x) \leq 0 \quad \forall i$

$\lambda_i \geq 0 \quad \forall i$

$\Rightarrow \lambda_i g_i(x) \leq 0$

$\Rightarrow \sum_{i=1}^{m} \bar{\lambda}_i g_i(x) \leq 0$

Sum is also equals $\underline{0}$

$$f(x) - f(\bar{x}) \geq -\sum_{i=1}^{m} \bar{\lambda}_i g_i(x) \geq 0$$

$$f(\bar{x}) \leq f(x) \quad \forall x \in S$$

$\underline{f(\bar{x}) \text{ is global minimum.}}$

Ex:

$$\text{Min } f = 2x_1 + x_2$$

$$\text{s.t} \quad x_1^2 + x_2^2 \le 4 \Rightarrow g_1 = x_1^2 + x_2^2 - 4 \le 0$$

$$x_1 - x_2 \le 0 \Rightarrow g_2 = x_1 - x_2 \le 0$$

Sol. First we will see the above Problem is a convex Programming problem (or) not, because then only the KKT conditions will be sufficient. Now it is a minimizing type problem and all constraints are less than equal to type. So it will be a convex programming problem where the function and all constraints are convex, so function is linear fn so it convex fn.

$f = 2x_1 + x_2 \hookrightarrow$ Linear so it is convex

Constraints $g_1 = x_1^2 + x_2^2 - 4 \le 0$

$g_2 = x_1 - x_2 \le 0$

Now first constraint g_1's hessian matrix is

$$\nabla^2 g_1 = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \quad \lambda_1 = 2, \lambda_2 = 2$$

The Eigen values of this matrix $\lambda_1 = 2, \lambda_2 = 2$

So: greater than 0

$\lambda_1 > 0, \lambda_2 > 0$

$\hookrightarrow$ positive definite that m

function is convex

Second constraints is the linear constraints is so its obviously convex function

KKT Conditions

Two KKT multiplier involved

Gradient x. Having two KKT constraints

$$\nabla f(x) + \lambda_1 \nabla g_1(x) + \lambda_2 \nabla g_2(x) = 0$$

$$\Rightarrow \underbrace{(2 \quad 1)}_{f} + \lambda_1 \underbrace{(2x_1 \quad 2x_2)}_{g_1} + \lambda_2 \underbrace{(1 \quad -1)}_{g_2} = (0 \quad 0)$$

$$\left.\begin{array}{l} \Rightarrow 2 + 2x_1\lambda_1 + \lambda_2 = 0 \\ \Rightarrow 1 + 2x_2\lambda_1 - \lambda_2 = 0 \end{array}\right\} \text{obtained from above}$$

Now what are the other conditions.

$$\left\{\begin{array}{l} \lambda_1 g_1(x) = 0 \Rightarrow \lambda_1 (x_1^2 + x_2^2 - 4) = 0 \\ \lambda_2 g_2(x) = 0 \Rightarrow \lambda_2 (x_1 - x_2) = 0 \end{array}\right.$$

From this obtained 4 cases $\lambda_i = 0$ / $(x$

Feasibility conditions

$$x_1^2 + x_2^2 - 4 \leq 0$$

$$x_1 - x_2 \leq 0$$

$$\lambda_1, \lambda_2 \geq 0$$

So these are cyclotic conditions.

Now solve these conditions Let us see

Case I : $\lambda_1 = \lambda_2 = 0$

Case II : $\lambda_1 = 0, \quad x_1 = x_2$

III : $x_1^2 + x_2^2 = 4, \quad \lambda_2 = 0$

IV : $x_1^2 + x_2^2 = 4, \quad x_1 - x_2 = 0$

Case 1: Now suppose the first condition

$\lambda_1 = \lambda_2 = 0$.

if $\lambda_1 = \lambda_2 = 0$ then

from the following two conditions

$$\left.\begin{array}{l} 2 + 2x_1\lambda_1 + \lambda_2 = 0 \\ 1 + 2x_2\lambda_1 - \lambda_2 = 0 \end{array}\right\} \Rightarrow \begin{array}{l} 2 + 2x_1(0) + (0) = 0 \Rightarrow \boxed{2 = 0} \\ \Rightarrow 1 + 2x_2(0) - (0) = 0 \\ \boxed{1 = 0} \end{array}$$

Hence $2 = 0$ (or) $1 = 0$ which is not possible,

So this condition is not possible.

So Case 1: Not possible.

Case 2:

$\boxed{\lambda_1 = 0}$ $\boxed{x_1 = x_2}$

$$\left.\begin{array}{l} 2 + 2x_1\lambda_1 + \lambda_2 = 0 \\ 1 + 2x_2\lambda_1 - \lambda_2 = 0 \end{array}\right\} \Rightarrow \begin{array}{l} 2 + 2x_1(0) + \lambda_2 = 0 \Rightarrow \boxed{\lambda_2 = -2} \\ 1 + 2x_2(0) - \lambda_2 = 0 \\ \boxed{\lambda_2 = 1} \end{array}$$

λ_2 having only one value here is non negative.

Case 2: Not possible.

Case III:

$x_1^2 + x_2^2 = 4$, $\lambda_2 = 0$

$2x_1\lambda_1 = -2$

$x_1\lambda_1 = -1$

$$\left.\begin{array}{l} 2 + 2x_1\lambda_1 + \lambda_2 = 0 \\ 1 + 2x_2\lambda_1 - \lambda_2 = 0 \end{array}\right\} \Rightarrow \begin{array}{l} 2 + 2x_1\lambda_1 + (0) = 0 \\ 1 + 2x_2\lambda_1 - (0) = 0 \end{array}$$

$x_1\lambda_1 = -1 \Rightarrow x_1 = -1/\lambda_1$

$x_2\lambda_1 = -1/2$

$\Rightarrow x_2 = \frac{-1}{2\lambda_1}$

$$\boxed{\begin{array}{l} x_1\lambda_1 = -1 \\ x_2\lambda_1 = -1/2 \end{array}}$$

$x_1^2 + x_2^2 = 4$

$\Rightarrow$

$x_1^2 + x_2^2 = 4$

$(-1/\lambda_1)^2 + \left(\frac{-1}{2\lambda_1}\right)^2 = 4$

$\frac{4+1}{4\lambda_1^2} = 4 \Rightarrow 5 = 16\lambda_1^2$

$\Rightarrow \lambda_1 = \pm\frac{\sqrt{5}}{4}$

$\boxed{\lambda_1 = \frac{\sqrt{5}}{4}}$

$\frac{-1}{\cancel{2}\frac{\sqrt{5}}{\cancel{4}2}} = \frac{-2}{\sqrt{5}}$

$\therefore \quad x_1 = \frac{-1}{\lambda_1} \Rightarrow \boxed{\frac{-4}{\sqrt{5}}}$

$x_2 = \frac{-1}{2\lambda_1} \Rightarrow \boxed{\frac{-2}{\sqrt{5}}}$

Now $x_1^2 + x_2^2 - 4 \leq 0$ ✓ hold.

satisfying.

Case IV.

Now $x_1 - x_2 \leq 0$

what is the x_1 & x_2

$\frac{-4}{\sqrt{5}} + \frac{2}{\sqrt{5}} \Rightarrow \frac{-4+2}{\sqrt{5}} \Rightarrow \frac{-2}{\sqrt{5}}$ which ≤ 0

So yes this condition also holds hence this point all the KKT condition.

Ex 2

$\text{Min } f: x_1^2 + x_2^2 - 2x_1$

$\text{s.t } g_1 = x_1^2 + x_2 - 1 \leq 0$